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The role of language teachers' perceptions and attitudes in ICT integration in higher education EFL classes in China

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To successfully adopt and implement information and communication technology (ICT) in language teaching and learning processes, it is essential to understand how and why language teachers use ICT and identify the barriers they encounter when attempting to incorporate ICT into their lessons. This study examined English as a foreign language (EFL) teachers' attitudes at the tertiary level towards integrating ICT in their classes based on technology acceptance model (TAM) factors using a questionnaire consisting of both closed- and open-ended questions. Data were collected from 68 EFL teachers at three institutions in Guangzhou, southern China. The findings validated the efficacy of the TAM components in assessing the integration of ICT in higher education EFL classes in China based on teachers' perceptions and attitudes and revealed teachers' positive attitudes and significant tendencies towards ICT integration in English classes. However, teachers' digital literacy and pedagogical and technological skills were found to be insufficient to successfully integrate ICT into tertiary English education. Most participants expressed a desire for in-service ICT training. Moreover, various obstacles were found to affect ICT integration in higher education English classes in China, including personal, institutional, and technological barriers.

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Introduction

The need to implement technology in education has dramatically increased worldwide. Information and communication technology (ICT) is becoming increasingly significant and essential in both educational systems and our daily lives (Demirci 2009), and the advent of novel technologies and digital resources in recent decades has profoundly affected educational landscapes and learning opportunities (Moradi and Chen 2019). Avidov-Ungar and Amir (2018) posited that technological advancements fundamentally alter societal structures and employment, while Chun (2019) highlighted that we have been trying to determine how education and literacy have changed due to technological development since the beginning of the twenty-first century.

ICT in education encompasses various forms — from basic presentation software to more sophisticated applications. The landscape of ICT integration in language education has evolved significantly, particularly during and after the COVID-19 pandemic. Recent research has identified several key themes in ICT education: (1) the increasing role of gamification and game-based learning in language acquisition; (2) the impact of informal, extramural English exposure through digital platforms; and (3) the shifting balance between classroom and online learning environments. Studies have shown that exposure to English through gaming, social media, other digital platforms, and blended learning environments significantly impacts learning outcomes (Calafato and Clausen 2024; Rød and Calafato 2023; Yang et al. 2020). For example, Yang et al. (2020) highlighted that adaptive gaming-based learning systems can significantly enhance learning outcomes when properly implemented. Similarly, Calafato and Clausen (2024) demonstrated how strategic ICT integration can support learning objectives. These findings underscore the importance of considering not only technology adoption but also implementation strategies when examining teachers' ICT use.

According to Ardiç and Çiftçi (2019), ICT refers to various types of technologies that help users access information through telecommunication. Additionally, as Hu and McGrath (2011) noted, ICT integration in educational settings has been a widespread practice for decades. Teachers play a pivotal role as primary agents in implementing innovative practices within the educational sphere (Fritz et al. 1995). Therefore, it is essential that teachers remain up to date on new educational technologies and enhance their technological literacy and pedagogical viewpoints regarding the adoption of these modern technologies in their teaching practices.

However, the extent to which teachers can adopt and adapt to these technologies often depends on the cultural and institutional frameworks in which they operate. Technological shifts have manifested differently across educational contexts; while Western educational systems have generally emphasised individual teacher autonomy in ICT adoption (Alonzo et al. 2024; Salokangas et al. 2019; Tondeur et al. 2017), Asian educational systems, particularly in China, have taken a more systematic, institution-driven approach (Chen et al. 2021; Liu et al. 2017b).

Previous research has extensively examined ICT adoption, highlighting its potential to engage students and promote autonomous learning (Chambers and Bax 2006; Liu et al. 2024; Muth and Lüftenegger 2024) while also underscoring persistent challenges, such as teachers' limited ICT skills, inadequate support structures, and difficulties in material selection (Barikzai et al. 2024; McGrath 2007; Wang 2007). Overall, the extant literature suggests a complex interplay of factors influencing successful ICT integration, which this study aims to further explore.

Rapid progress in ICT has led to impressive and considerable changes in education and educational systems, and language

teachers have had to cope with the challenges of this technological development in education alongside ongoing pedagogical reforms. According to Ardiç and Çiftçi (2019), the utilisation of ICT in language learning and teaching demands more meticulous planning and design compared to other disciplines. In leveraging the potential of cutting-edge technologies, teachers fulfil a critical function by adapting learning materials and activities to meet their students' specific needs (Aşık et al. 2020). While this practice aligns with global trends in ICT adoption, studies also indicate significant regional and contextual variations in teachers' readiness and ability to integrate ICT effectively. For instance, Fraillon et al. (2014) observed a general increase in teachers' confidence with ICT applications over time but noted that this did not necessarily result in frequent or efficacious ICT use in classrooms. Similarly, research by Alkhalwaleh and Menchaca (2014) and Ekberg and Gao (2018) revealed that several barriers, such as insufficient training and institutional support, continue to impede teachers' proactive utilisation of ICT. These findings highlight the importance of examining contextual factors and teacher perceptions, particularly in understudied disciplines such as EFL.

Previous studies (e.g., Estrada and Prasolova-Førland 2022; Liu et al. 2017b; Lv et al. 2024; Peng et al. 2023) have identified gaps between the available ICT resources and their actual implementation in classrooms. However, while these studies established the existence of adoption barriers, they did not fully explore how institutional support mechanisms influence technology acceptance, particularly in Chinese higher education EFL contexts. This study seeks to address this gap and identify the distinct challenges EFL teachers face in integrating technology for language instruction, such as balancing linguistic and technological competencies.

Despite the potential advantages of ICT integration in language education and the presence of ICT resources, educators' proactive utilisation of ICT remains infrequent, as indicated by reports from the European Commission (2013) and studies by Chen et al. (2021) and Peng et al. (2023). In this context, the unprecedented shift to online learning brought about by the COVID-19 pandemic has accelerated the integration of ICT in education while exposing substantial disparities in digital readiness. For instance, studies by Chen et al. (2021) and Wu et al. (2019) reported that rural Chinese schools, in stark contrast to better-equipped urban institutions, often face insufficient Internet infrastructure and lack access to updated digital tools. This transformation has particularly impacted EFL instruction, where traditional face-to-face interaction has been considered crucial for language acquisition. The pandemic has underscored the need for a more nuanced understanding of how teachers adapt to and implement ICT-based teaching strategies in language education.

The expansion of ICT's role in education during this period has made it more critical than ever to examine how teachers perceive and utilise technologies, especially as digital learning continues to be a pivotal component of the post-pandemic educational landscape. Thus, this study, which explores EFL teachers' attitudes towards ICT integration at the tertiary level in China through the lens of the TAM, is both timely and relevant. By investigating teachers' perceptions of ICT's usefulness, ease of use, and institutional support, this study aims to address the ongoing need for sustainable digital teaching practices beyond the pandemic. Our findings enhance the broader dialogue on how digitalisation in education can evolve and offer insights into both the opportunities for innovation and persistent challenges that need to be addressed. A crucial factor in technology adoption that is particularly relevant to this study is perceived ICT support, namely, the extent to which teachers believe they have access to necessary technological resources and assistance. This study addresses the

underexplored construct of perceived ICT support in Chinese higher education EFL settings, which is a critical variable influencing technology acceptance patterns.

While extensive research exists on educators' technology adoption, this study offers several unique contributions. First, it specifically examines the intersection of perceived ICT support and technology acceptance in Chinese higher education EFL contexts, an area that, thus far, remains underexplored. Second, it extends the TAM by incorporating the construct of perceived ICT support, which holds particular relevance for Chinese educational settings. Third, it provides timely insights into how the post-pandemic transition has shaped EFL teachers' engagement with ICT in higher education.

Literature review

The relationship between attitudes and behaviours has been the primary focus of education and social psychology research for many years. Researchers have continuously investigated whether attitudes prompt specific types of behaviours. Baker (1992) defined *attitude* as a hypothetical construct used to explain the orientation and consistency of human behaviour. Various studies examining the relationship between attitudes and behaviours (see Gwinn and Krajbich 2020; Moradi and Chen 2022; Moradi and Gupta 2017; Passafaro 2020) consider attitudes to be influential factors shaping individuals' conduct in their everyday lives. Gwinn and Krajbich (2020) underscored the pivotal function of attitude in routine decision-making processes, contending that various facets of attitudes exert substantial influence on the manner in which decisions are formulated. Conversely, other researchers (see Karahan 2007; Luttrell and Sawicki 2020) have raised doubts about the connection between attitudes and behaviours, suggesting that attitudes may not consistently predict actions and may have minimal or no correlation.

While various factors contribute to the effective integration of ICT into higher education, teachers' attitudes and technological competencies undoubtedly serve as cornerstones (Yang et al. 2023). In the context of education, teachers' attitudes towards technology play a decisive role in their willingness to adopt and integrate ICT into their teaching practices. Studies have demonstrated that positive attitudes towards ICT are strongly associated with higher levels of adoption (Krejins et al. 2013; Yang et al. 2023). However, various barriers, such as inadequate institutional support, lack of training, and low digital literacy, often hinder teachers from fully utilising ICT (Ertmer et al. 2012; Eze et al. 2021; Nii Akai Netey et al. 2024). While these findings underscore the importance of teacher attitudes, they also reveal gaps in our understanding of how external factors, such as perceived ICT support, influence these attitudes.

Several investigations into teachers' attitudes regarding technology have employed the TAM, originally introduced by Davis (1986), to depict how individuals accept and engage with technology. The TAM posits that two key factors influence users' perspectives, choices, and adoption of a new technology: 1) 'perceived usefulness', which refers to an individual's belief that employing a certain system will improve their work performance, and 2) 'perceived ease of use', which denotes the belief that utilising the system will be effortless (Davis et al. 1989). The TAM is a comprehensive framework that incorporates both attitudes and behaviours concerning ICT. However, this study concentrates on aspects of the TAM related to perceptions and attitudes.

The evolution of the TAM in educational research has seen various modifications and extensions to address context-specific needs. While the original TAM (Davis 1986) focused primarily on business settings, subsequent adaptations for educational contexts have incorporated facilitating conditions and technological

complexity (Teo 2019) and emphasised the role of professional development support (Scherer et al. 2019). These modifications underscore the importance of contextualising the TAM for specific educational settings while preserving its core theoretical framework.

The TAM has been extensively evaluated in educational settings, and its effectiveness has been substantiated (Legris et al. 2003), with numerous investigations demonstrating that attitude significantly influences individuals' intentions to utilise technology (e.g., Baş et al. 2016; Krejins et al. 2013; Peng et al. 2023; Shin et al. 2014; Yang et al. 2023). Jeyaraj et al. (2006) examined 11 technology adoption frameworks across 99 relevant studies and found that the TAM was the most frequently employed model for individual information technology (IT) adoption. Their review also revealed that perceived usefulness and ease of use were the most commonly used predictors of technology adoption, with the former considered one of the most dependable indicators.

While the TAM has demonstrated robust predictive power across various educational contexts, its application in non-Western settings, particularly in Chinese higher education, necessitates careful consideration of cultural and institutional factors. In particular, the model's core variables of perceived usefulness and ease of use may be influenced differently in collectivist cultures where institutional harmony and group consensus often play crucial roles in technology adoption decisions (Liu et al. 2017a). Meta-analyses have revealed that while perceived usefulness consistently emerges as a strong predictor in Western contexts, studies in Asian educational settings suggest that social influence and institutional support may play more significant roles (Scherer et al. 2019).

Nevertheless, despite the TAM's predictive power, critics have questioned its adequacy in capturing the complexity of teaching contexts. Researchers such as Bagozzi (2007) and Teo (2019) argue that the TAM's simplicity, while valuable for prediction, may oversimplify the dynamic nature of educational technology adoption. This is particularly true in non-Western contexts where cultural factors, such as power distance and collectivism (see Hofstede 2001), may mediate the relationships between the TAM constructs differently than in Western settings (Teo 2019).

Recent critiques of the TAM in educational research (e.g., Al-Nuaimi and Al-Emran 2021; Teo 2019) have emphasised the need to consider contextual variables beyond the original model's scope. In Chinese higher education, where institutional support structures and cultural values significantly influence teaching practices, the traditional TAM framework requires modification to better reflect local realities. For EFL teachers in particular, technology adoption decisions are shaped not only by individual perceptions but also by departmental policies, peer influence, and cultural attitudes towards foreign language instruction.

Although the TAM has been widely validated, relatively few studies have integrated teachers' perceived ICT support as a primary predictor of technology adoption models or explored the connections among perceived ICT support, teachers' attitudes towards technology, and their intention to utilise it within the TAM framework. This gap in the research presents an opportunity to broaden the model by including these crucial elements. In the existing TAM structure, an individual's stance on ICT is influenced by their perception of its usefulness and ease of use. However, this attitude and subsequent behavioural intentions may also be connected to underlying perceptions of ICT support. Consequently, the present study explores how teachers' perceived ICT support impacts their assessment of ICT's usefulness and ease of use, as well as their attitudes towards and intention to employ it. This focus is particularly relevant for EFL teachers, whose technology adoption decisions depend heavily on the institutional support and resources available within their teaching contexts.

The integration of perceived ICT support into the TAM represents a crucial theoretical extension, thereby addressing a key limitation of the original model. Importantly, this addition acknowledges that technology adoption in educational settings is not merely an individual decision but is embedded within institutional and social support systems. The relationship between perceived support and technology acceptance is particularly pronounced in hierarchical educational systems such as those found in China, where institutional backing significantly influences individual teacher decisions. For EFL teachers, whose classrooms often require technology to facilitate interactive language learning and access to global resources, institutional support can play a decisive role in determining the success of ICT integration (Cukur 2023; Hasumi and Chiu 2024).

While it has been largely pointed out that teachers' effective use of ICT requires adequate institutional support (Ertmer et al. 2012; Eze et al. 2021; Liu et al. 2017a), the dynamic nature of this relationship warrants renewed examination, particularly in the Chinese higher education EFL context. The hierarchical structure of Chinese educational institutions, combined with the unique challenges of language teaching and recent post-pandemic transitions, creates a distinct environment where the influence of perceived support on technology adoption may manifest differently. For instance, EFL teachers in these settings may rely on institutional support not only for access to technology but also for professional development opportunities, such as training in language-specific digital tools or platforms. Furthermore, the rapid evolution of educational technologies and changing student needs in language learning necessitate fresh investigation into how institutional support mechanisms affect teachers' technology acceptance patterns.

Perceived ICT support refers to the extent to which users believe they can access assistance and resources related to technology use. This support can come from various sources, such as IT helpdesks, technical documentation, user training, peer assistance, and organisational policies promoting technology use. As an additional primary variable in the TAM, perceived ICT support helps capture the influence of external factors on users' attitudes and intentions towards technology adoption. Higher perceived support is likely to positively impact perceived usefulness and ease of use as it helps users overcome potential obstacles and enhances their confidence in effectively using technology. In the context of EFL education, this support may include training in language-specific applications, access to digital teaching aids, or peer mentorship programmes to integrate ICT into language instruction. However, few studies have incorporated 'perceived ICT support' as one of the main predictors of models focused on technology adoption. Additionally, these models have often been tested without adequately considering teachers' academic discipline (Liu et al. 2017a).

Although previous research indicates that the cultural context of a subject area can significantly impact the acceptance or rejection of technological tools, such as computers and the Internet, within educational institutions (Cheng et al. 2022; Dressman et al. 2023; Fraillon et al. 2014), more research explicitly targeting teachers in specific subjects, such as EFL, is needed. Given that EFL teachers often operate at the intersection of institutional and cultural factors, understanding how these dynamics shape their technology-related behaviours could offer valuable insights for designing tailored ICT support mechanisms. Such research would contribute to a deeper understanding of teachers' technology-related behaviours and offer practical guidance for institutions seeking to enhance ICT integration in specific disciplines.

ICT research in the context of China

Previous studies have explored ICT policy and its integration in Chinese education from various perspectives, highlighting

distinct challenges shaped by institutional, cultural, and pedagogical factors. Unlike Western contexts, where teacher autonomy often drives ICT adoption, Chinese educational institutions typically employ a more systematic, top-down approach. This approach, combined with the specific demands of EFL instruction and post-pandemic shifts, creates a unique context for ICT implementation. Despite generally positive institutional attitudes, a significant gap exists between policy expectations and actual practice. Recent studies (e.g., Chen et al. 2021; Huang 2022) have shown that while Chinese EFL teachers express positive attitudes towards ICT, their usage of ICT often remains limited to basic applications.

Existing research on ICT integration in Chinese education reveals three key trends: (1) a predominant focus on infrastructure and policy rather than teacher perspectives (e.g., Wu et al. 2019; Zeng 2022); (2) a concentration on K – 12 settings, leaving higher education relatively underexplored (e.g., Cai 2016; Deng et al. 2014; Wu et al. 2019); and (3) an emphasis on technical rather than pedagogical aspects of ICT use (e.g., Hu and McGrath 2012). These gaps underscore the need for research on EFL teachers' perspectives in higher education, where pedagogical and institutional contexts differ markedly from K – 12 settings. Zeng (2022) reviewed China's policy framework for integrating ICT into education from 1988 to 2021 and identified key phases in policy evolution, including infrastructure development, teaching innovation, and ICT competence cultivation. These findings suggest that effective policy implementation is crucial for enhancing teachers' acceptance of ICT by providing the necessary resources and training. Wu et al. (2019) emphasised the need for teacher-centred approaches, identifying gaps in training and design-thinking skills. Similarly, Hu and McGrath (2012) mentioned inadequate ICT skills and insufficient technical support as major barriers in EFL classrooms.

Studies such as Deng et al. (2014) and Li (2015) revealed the intersection of pedagogical beliefs and ICT usage, highlighting the importance of constructivist approaches and curriculum design. Huang (2022) proposed innovative frameworks for integrating ICT into university-level English instruction and illustrated its potential to enhance teaching outcomes and student learning dispositions. Similarly, Li (2022) demonstrated the efficacy of mobile-assisted learning tools in improving language skills such as fluency and accuracy in spoken English, and Cai (2016) and Zhang (2012) studied the integration of ICT into primary school curricula. All of these studies supported ICT integration in schools while underscoring the important role of schools in providing the necessary educational infrastructure and encouraging teachers to effectively use modern technologies in their teaching. Huang (2022) proposed innovative frameworks for integrating ICT into university-level English instruction while demonstrating its potential to enhance teaching outcomes and student learning dispositions. Similarly, Li (2022) illustrated the efficacy of mobile-assisted learning tools in improving language skills, such as fluency and accuracy in spoken English.

To date, most ICT research in China has focused on K – 12 education, leaving higher education, particularly EFL contexts, relatively underexplored. This study addresses this gap by investigating the factors that influence ICT adoption among university-level English teachers. By examining teachers' perceptions using the TAM framework, this study aims to uncover the motivational and contextual factors that are essential for effective ICT integration in higher education classrooms.

While the TAM has been extensively validated in Western contexts (Jeyaraj et al. 2006; Viberg et al. 2024), its application in non-Western settings, especially in Chinese EFL higher education, remains limited. Moreover, most TAM studies focus on students while neglecting how teachers' perceptions, attitudes,

and institutional support influence ICT adoption. An important limitation of the existing TAM research is its limited incorporation of contextual and external factors, such as teachers' perceived ICT support. Although several studies (e.g., Ertmer et al. 2012; Eze et al. 2021) have demonstrated that institutional support can significantly enhance teachers' confidence and willingness to adopt ICT, this variable is rarely integrated into the TAM framework. This study incorporates perceived ICT support as a key variable and explores how it influences teachers' attitudes and intentions to adopt ICT in their teaching practices.

ICT integration in EFL classes involves balancing technological tools with the interactive nature of language learning. While some studies broadly address ICT implementation across a wide range of disciplines, they have largely overlooked the unique pedagogical challenges EFL teachers face, particularly in the culturally and institutionally distinct environment of Chinese higher education.

Although ICT research has extensively examined infrastructure and student outcomes (e.g., Zeng 2022; Li 2022), less attention has been paid to the interplay between teachers' psychological readiness, institutional support, and pedagogical beliefs. This study addresses these gaps and explores how cultural and institutional contexts in Chinese higher education shape EFL teachers' adoption behaviours using the TAM framework to analyse barriers and motivations systematically.

This gap is particularly evident in the context of Chinese higher education, where national policies emphasise ICT integration, but teachers' adoption remains uneven (Huang 2022; Zeng 2022). Thus, a more nuanced understanding of how EFL teachers perceive and utilise ICT is needed. Additionally, while studies have identified barriers to ICT integration, such as inadequate training and lack of support (Ertmer et al. 2012; Peng et al. 2023), few have systematically examined how they interact with teachers' attitudes and perceptions of ICT support within the TAM as an established theoretical framework. This study seeks to address these gaps by focusing on EFL teachers in Chinese higher education, a context that has received relatively little attention in ICT research. The research questions explored in this study are as follows:

RQ1: How well do the TAM factors — teachers' perception of the usefulness of ICT, support, perceived ease of use, and ICT attitude—predict English language teachers' intentions to use ICT in the future?

RQ2: What factors affect the adoption and integration of ICT in language education, particularly in English language classes, from the teacher's perspective?

Method

Participants. To investigate the research questions, a structured questionnaire was created and sent to 186 randomly selected EFL teachers employed at the tertiary level at three institutions in Guangzhou, a first-tier city in southern China. Of the three selected institutions, one holds the status of a key national university, another is classified as a key provincial university, and the third is designated as a four-year college. These three institutions were strategically selected to represent the hierarchical diversity within China's higher education system. This purposive sampling approach ensures coverage of different institutional contexts, resource levels, and teaching environments, thus allowing for a more comprehensive understanding of ICT integration across various tertiary settings. The inclusion of institutions from different tiers provides insights into how institutional status and resource availability may influence teachers' technology acceptance patterns. These establishments are categorised into separate tiers within China's higher education ranking system and receive

varying levels of preference and favouritism regarding resource allocation, teaching faculty, and student quality (Wu 2017). The respondents consisted of 142 female ($N = 142$, 76.5%) and 44 male ($N = 44$, 23.5%) EFL teachers ranging from 25 to 48 years of age. The respondents' teaching experience ranged from 2 to 18 years. All participants had a master's degree related to EFL and applied linguistics, ($N = 186$) and 18 had doctoral degrees. All participants ($N = 186$) reported having online teaching experience, which was attributed to the emergence of the COVID-19 pandemic and the mandates of their respective institutions to provide online instruction during the outbreak.

It is important to note that the universal online teaching experience among participants, necessitated by the COVID-19 pandemic, represents a unique contextual factor that may have influenced their attitudes towards ICT integration. This forced transition to online instruction provided all participants with intensive exposure to educational technology, potentially shaping their perceptions of ICT usefulness, ease of use, and support needs differently than what might be observed in a non-pandemic context. This contextual consideration should be kept in mind when interpreting the study's findings as it represents both a limitation and an opportunity to understand technology acceptance during a period of rapid educational transformation.

Instrument. Data were collected from participants via a structured questionnaire using the electronic tool Qualtrics, which was developed based on the TAM proposed by Davis et al. (1989). As Fishman (1970) mentioned, there are three main approaches to determining and extracting responses from participants: questionnaires, interviews, and observations. In the questionnaire approach, participants respond to two types of questions: open-ended and closed. In this research, we used both types.

Apart from the initial section, which was incorporated to collect the respondents' background information as reported earlier, the questionnaire comprised one open-ended question and 39 statements that participants were asked to agree or disagree with on a 5-point Likert scale, where 1 indicated they 'strongly disagreed' and 5 indicated that they 'strongly agreed' with the given statements (1 = strongly disagree, 2 = disagree, 3 = neither agree nor disagree, 4 = agree, and 5 = strongly agree). It should be noted that for the data analysis and presentation of results, the positive responses for both the 'agree' and 'strongly agree' categories were combined and presented as 'agree', while the negative ones for 'disagree' and 'strongly disagree' were combined and presented as 'disagree'. The questionnaire items were adapted from previous research and literature on ICT integration in education (e.g., Chiou 2011; Liu et al. 2017a; Liu et al. 2019) and were modified and developed based on the TAM. The adaptation of items from previous research was strategically chosen for three reasons: (1) Liu et al. (2017a, 2019) demonstrated strong reliability and validity in Chinese educational contexts, while Chiou's (2011) validated TAM measures provided complementary insights from a broader educational perspective; (2) collectively, these studies effectively captured the nuanced aspects of technology acceptance in educational environments, particularly in language teaching settings; and (3) their validated measures allowed for meaningful comparisons with the existing literature while maintaining contextual relevance to Chinese higher education through appropriate cultural and linguistic adaptations. The 39 statements were divided into five categories, with 12 statements relating to 'perceived usefulness', ten statements relating to 'perceived ICT support', nine statements relating to 'perceived ease of use (ability)', four statements relating to 'attitudes towards ICT use', and four statements relating to 'behavioural intention to use ICT'. The items were

entered into IBM SPSS Statistics software version 25, and the output for Cronbach’s alpha, as illustrated in Table 1, revealed the reliability of the questionnaire.

Procedure. This study employed an embedded mixed-methods design, where qualitative data (open-ended responses) were nested within a predominantly quantitative framework (structured questionnaire). This design was chosen because it enables a comprehensive understanding of technology acceptance through quantitative measurements while simultaneously capturing nuanced personal experiences through qualitative responses. Moreover, this approach facilitates triangulation of findings through multiple data sources. Finally, the qualitative component allows for deeper exploration of unexpected quantitative results, providing richer insights into teachers’ technology acceptance patterns. Prior to distributing the questionnaires, the researchers provided the necessary information to the participants. To enhance its legitimacy, the questionnaire, which is the main instrument used in this research, was given to colleagues and experts in the field to evaluate and confirm its validity. A pilot study was also conducted with 18 English teachers to test the methods used in the final survey and to confirm whether the questions were clear and understandable to the participants. The questionnaire was modified in response to the pilot study and expert feedback. Modifications to the questionnaire were found to be useful in terms of logical sequencing, clarity, and meaningful analysis of the responses obtained. The data collection period, which occurred after extensive pandemic-related online teaching experience, provides a unique perspective on technology acceptance in higher education. This context may have influenced participants’ responses and should be considered when interpreting the results within the TAM framework.

Data analysis. Multiple statistical procedures were employed to fully address the research questions. Descriptive statistics were calculated for demographic information and individual item responses, and Cronbach’s alpha coefficients were computed to assess the internal consistency reliability of each construct. To examine the relationships between variables, we performed a

correlation analysis that determined the associations between perceived usefulness, ICT support, ease of use, attitudes towards ICT, and behavioural intention. Additionally, we conducted multiple regression analysis to assess the predictive power of the independent variables on behavioural intention to use ICT. One-sample t-tests were conducted to evaluate the significance of mean differences, and ANOVA was employed to examine the variance explained by the predictor variables.

The qualitative data derived from the open-ended question were analysed utilising thematic analysis adhering to Braun and Clarke’s (2006) six-step approach: familiarisation with data, generation of initial codes, identification of themes, review of themes, definition of themes, and production of the report. Two researchers independently coded the responses to ensure coding reliability, with disagreements resolved through discussion until a consensus was reached. For the integrated analysis, the qualitative findings were used to elaborate and explain the quantitative results, thus providing a more comprehensive understanding of teachers’ technology acceptance patterns.

Theoretical framework

The theoretical framework of the present study is based on the TAM, which was specifically created to describe individual technology acceptance and to be used globally with different computer technologies and in different organisational contexts (Davis et al. 1989). As aforementioned, the TAM presumes that two specific beliefs, ‘perceived usefulness’ and ‘perceived ease of use’, are paramount in influencing computer acceptance behaviour. Legris et al. (2003) pointed out that although the TAM has been a useful and practical model for analysing ICT usage, it also requires two variables — ‘perceived usefulness’ and ‘perceived ease of use’ — as essential determining factors to predict behavioural intention to utilise technological innovations. This study modifies the TAM by including ‘perceived ICT support’ as another key variable for ICT integration (See Fig. 1). In line with the research questions, this model was adopted for the present research to provide a conceptual and theoretical basis, as well as a clear description of how tertiary-level EFL teachers in China perceive ICT integration in language education.

Table 1 Reliability of the instrument.		
	Number of items	Cronbach's alpha
PU	12	0.841
PICTS	10	0.811
PEOU	9	0.822
ATU	4	0.796
Intention to use ICT	4	0.804

PU perceived usefulness, PEOU perceived ease of use, PICTS perceived ICT support, ATU attitudes towards ICT, BI behavioural intention, AU actual ICT use.

Results

To address the research questions, we conducted various statistical analyses to acquire data, including sample t-tests and descriptive statistics for each variable, the results of NOVA analysis, and coefficients from regression analysis.

As illustrated in Table 2, participants reported positive perceptions of and attitudes towards ICT. The analysis of descriptive statistics revealed consistently positive perceptions towards ICT integration across all TAM components. Notably, perceived usefulness emerged as the most positively rated factor, suggesting teachers recognise the value of technology in their teaching practice.

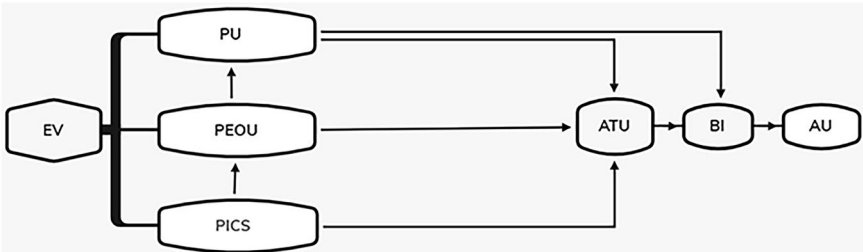


Fig. 1 Proposed model. EV External Variables, PEOU Perceived Ease of Use, PU Perceived Usefulness, PICTS Perceived ICT Support, ATU Attitudes towards ICT, BI Behavioural Intention, AU Actual ICT Use.

Table 2 Descriptive statistics for variables.

Variables	M
PU	4.811
PICTS	4.391
PEOU	4.022
ATU	4.181
BI	4.219

PU perceived usefulness, PICTS perceived ICT support, PEOU perceived ease of use, ATU attitudes towards ICT, BI behavioural intention.

Table 3 Correlations between variables.

Variables	Perc. usefulness	ICT support	Ease of use	Attitudes towards ICT use	Intention to use ICT
PU	1				
PICTS	0.186*	1			
PEOU	0.362**	0.587**	1		
ATU	0.561**	0.312**	0.367**	1	
BI	0.432**	0.428**	0.424**	0.458**	1

PU perceived usefulness, PICTS perceived ICT support, PEOU perceived ease of use, ATU attitudes towards ICT, BI behavioural intention.

* $p < 0.05$; ** $p < 0.01$.

Table 4 One-sample t-test.

T.V = 37					92% Conf. interval of difference	
	df	sig	t	Mean dif.	Lower	Upper
PU	67	0.000	32.068	12.645	11.521	13.455
PICTS	67	0.000	28.628	8.412	7.521	9.911
PEOU	67	0.000	12.377	11.154	10.218	12.841
ATU	67	0.000	20.506	11.419	9.819	12.261

PU perceived usefulness, PICTS perceived ICT support, PEOU perceived ease of use, ATU attitudes towards ICT.

Table 3 depicts the correlations between the variables, highlighting the positive relationships between the TAM factors.

The correlation analysis (see Table 3) revealed significant relationships between the TAM components, with particularly strong correlations between ICT support and ease of use ($r = 0.587$, $p < 0.01$), suggesting institutional support's crucial role in enhancing teachers' perceived ability to use technology effectively, and between perceived usefulness and attitudes towards ICT use ($r = 0.561$, $p < 0.01$), indicating that recognition of ICT benefits strongly influences teaching attitudes. These relationships suggest that institutional support and perceived benefits play crucial roles in shaping teachers' technology acceptance.

Table 4 presents the one-sample t-test results examining TAM factors' significance in predicting ICT integration intentions.

The t-test results revealed that all of the TAM components — perceived usefulness, ICT support, attitudes, and ease of use — were statistically significant predictors of ICT integration (all $p < 0.001$). Notably, perceived usefulness showed the strongest effect ($t = 32.068$), followed by ICT support ($t = 28.628$), suggesting these factors are particularly crucial in teachers' technology adoption decisions. The significant positive mean differences across all factors indicate that teachers' perceptions

Table 5 NOVA analysis.

Analysis model	Sig	Mean square	Sum of square	F	Df
Regression	0.000 ^a	6.338	19.344	61.315	2
Residual		0.146	10.769		65
Total			29.113		67

^aPredictors: (constant), perceived usefulness, support, ease of use, and ICT attitude.

Table 6 Coefficients analysis.

	B	Beta	t	Std. Err	Sig
PU	1.899	0.588	141.028	0.007	0.000
PICTS	0.932	0.452	102.873	0.011	0.000
PEOU	0.812	0.417	99.136	0.012	0.000
ATU	0.872	0.424	99.891	0.012	0.000

PU perceived usefulness, PICTS perceived ICT support, PEOU perceived ease of use, ATU attitudes towards ICT.

Table 7 Model regression sum.

			Ch. Statistics		
R	sig	R ² v	Adjusted R ² v	Std. Err.	df1
0.895 ^a		0.000	0.780	0.770	0.32617
58					6

^aPredictors: (constant), perceived usefulness, support, ease of use, and ICT attitude.

consistently exceeded the test value, thus demonstrating strong potential for technology acceptance in this educational context.

To examine how well TAM factors predicted teachers' intentions to use ICT, multiple regression analysis was conducted (see Table 5).

The regression model was statistically significant, $F(2,65) = 61.315$, $p < 0.01$, indicating that TAM factors collectively were strong predictors of behavioural intention to use ICT. The model's strength justified further examination of individual factor contributions through coefficient analysis.

The results of the coefficient analysis (see Table 6) revealed the relative influence of each TAM factor on teachers' behavioural intention to use ICT (all relationships significant at $p < 0.001$). Perceived usefulness emerged as the strongest predictor ($\beta = 0.588$), followed by ICT support ($\beta = 0.452$), attitudes towards ICT ($\beta = 0.424$), and perceived ease of use ($\beta = 0.417$). This hierarchical pattern suggests that teachers' technology adoption is primarily driven by their perception of ICT's utility in teaching, while institutional support plays a crucial secondary role. The consistent significance of all factors indicates that successful ICT implementation requires a comprehensive approach addressing both utility perceptions and support structures in higher education EFL contexts.

As shown in Table 7, the regression model demonstrated strong explanatory power, with an adjusted R^2 value of 0.77, indicating that the TAM factors collectively explained 77% of the variance in Chinese EFL teachers' behavioural intention to use ICT in higher education settings. This robust model fit ($R = 0.895$, $p < 0.001$) suggests that the combination of perceived usefulness, ICT support, ease of use, and attitudes offers a holistic

framework for understanding technology acceptance among EFL teachers in Chinese higher education institutions.

To complement the quantitative findings and gain deeper insights into implementation challenges, an open-ended question explored Chinese EFL teachers' perceived barriers to ICT integration. The analysis of the responses ($N = 186$) revealed three major themes affecting technology adoption in higher education settings:

– Professional development barriers

- Limited ICT experience ($f = 177$, $\% = 95.2$). 'Despite my 15 years of teaching experience, I still struggle with implementing new technologies effectively due to limited training opportunities. I also find myself struggling to keep up with rapidly evolving educational technologies. We need systematic training to bridge this gap'. [Teacher 23]
- Lack of pedagogical ICT training programmes ($f = 176$, $\% = 94.6$). 'The training we sometimes receive focuses on technical aspects, but we need guidance on how to effectively integrate these tools into our teaching methodology'. [Teacher 98]
- Digital literacy gaps ($f = 158$, $\% = 84.9$). 'I often feel overwhelmed by the complexity of new educational platforms. Basic digital skills training would help us feel more confident'. [Teacher 15]

– Institutional Constraints

- Workload and time pressure ($f = 172$, $\% = 92.5$). 'With 16 teaching hours weekly plus administrative duties, finding time to implement new technologies is extremely challenging'. [Teacher 36]
- Lack of appropriate educational ICTs ($f = 153$, $\% = 82.3$). 'The available technology often doesn't align with our curriculum needs. We need approved tools specifically designed for language teaching'. [Teacher 8]
- Restricted and inflexible curricula ($f = 148$, $\% = 79.6$). 'The rigid curriculum structure leaves little room for innovative teaching approaches. We're too focused on covering the prescribed content'. [Teacher 161]
- Inflexible structure of traditional educational system ($f = 142$, $\% = 76.3$). 'Our traditional assessment system doesn't accommodate technology-enhanced learning well. We need systemic changes'. [Teacher 19]

– Personal and Technical Barriers

- Limited self-efficacy ($f = 126$, $\% = 67.7$). 'I understand technology's importance but lack confidence in using it effectively. I worry about making mistakes in front of students'. [Teacher 47]
- Access limitations ($f = 118$, $\% = 63.4$). 'Access to reliable technology remains inconsistent. When we finally get comfortable with one system, either it breaks down or we lose access'. [Teacher 52]
- Confidence issues ($f = 104$, $\% = 55.9$). 'The fear of technical failures during class makes me hesitant to rely on technology for important lessons'. [Teacher 139]

These barriers reflect the unique challenges within Chinese higher education, where traditional teaching approaches intersect with technological innovation. The high frequency of professional development barriers (95.2% citing limited ICT experience) suggests a systematic need for reform in teacher training programmes, particularly in adapting to rapid technological changes while maintaining pedagogical effectiveness in EFL instruction.

The interconnected nature of these barriers reveals deeper systemic challenges. For instance, teachers' comments about workload pressures (92.5%) often intersect with their concerns about limited ICT experience (95.2%), suggesting that time constraints not only affect immediate implementation but also hinder long-term professional development. Similarly, the high percentage of teachers citing pedagogical ICT training needs (94.6%) alongside curriculum inflexibility (79.6%) indicates that technological integration challenges are compounded by structural educational constraints.

Teachers' narratives particularly emphasise how personal and institutional barriers reinforce each other. For example, limited self-efficacy (67.7%) appears to be closely linked to inadequate training opportunities, while concerns about technical failures (55.9%) reflect both personal confidence issues and institutional support limitations. These interconnections suggest that interventions must address both individual and systemic factors simultaneously to be effective.

The quantitative and qualitative findings complement each other in several ways. Notably, the strong influence of ICT support ($\beta = 0.452$) in the regression analysis aligns with the qualitative findings wherein institutional constraints (i.e. workload, curriculum restrictions, lack of appropriate ICTs) were frequently mentioned barriers (92.5%, 79.6%, and 82.3% respectively). While perceived usefulness emerged as the strongest predictor ($\beta = 0.588$) in the quantitative analysis, the qualitative data revealed that teachers face significant implementation challenges despite recognising ICT's value, particularly in terms of training needs (94.6%) and digital literacy gaps (84.9%).

Discussion

RQ1. The results provide empirical evidence that supports the effectiveness of the TAM in elucidating the intentions of Chinese EFL teachers regarding technology adoption and confirm the effectiveness of the TAM components in assessing ICT integration in higher education EFL classes in China based on teachers' perceptions and attitudes. Specifically, the TAM factors have significant and positive relationships with each other, and EFL teachers' attitudes towards ICT, their intention to use it in their classes, and/or their actual ICT use can be determined' from teachers' perceptions of ICT usefulness, ICT support, and their ability or how easy it is to use. Furthermore, the teachers' attitudes towards ICT had a considerable effect on their intentions to use ICT in their classes. However, these findings need to be understood within China's unique educational landscape, where the integration of ICT represents not just a technological shift but a fundamental transformation of pedagogical traditions. Unlike Western educational contexts, where individual teacher autonomy often drives innovation, Chinese higher education operates within a more structured framework that emphasises collective advancement and institutional harmony.

These findings align with those of earlier studies that confirmed the effectiveness of the TAM in examining teachers' perceptions and attitudes towards ICT, their intentions to use it, and actual ICT use in diverse educational settings (Chen et al. 2021; Haines 2015; Legris et al. 2003; Liu et al. 2017a; Turner et al. 2010). In their investigations of pedagogical beliefs and attitudes toward ICT among EFL teachers in China, Liu et al. (2017a) and Cao et al. (2023) arrived at comparable outcomes, with their findings highlighting that teachers' perspectives on ICT are markedly shaped by their perceptions of its usefulness and ease of use. This alignment with previous research can be attributed to the similar cultural and institutional contexts within Chinese higher education, where traditional teaching approaches are gradually being transformed by technology integration demands.

The consistency in findings suggests that, despite unique and educational contexts, Chinese EFL teachers' technology adoption patterns follow universal TAM principles.

Furthermore, the results indicate a substantial connection between teachers' attitudes towards ICT and their intentions to incorporate it into their practices. As noted by Shahzad et al. (2024), teachers who perceive strong benefits of ICT exhibit a greater tendency to accept and implement it in their lessons. This relationship between perceived benefits and adoption intention appears particularly robust in our study, possibly because Chinese higher education institutions have recently emphasised technology integration in their strategic planning, creating an environment where teachers are more likely to act on positive perceptions of ICT usefulness. Consistent with our findings, Peng et al. (2023) emphasised that perceived ease of use plays a critical role in ICT adoption. When teachers believe that integrating ICT into their teaching requires minimal effort, they are significantly more likely to embrace it (Peng et al. 2023).

In this study, the TAM was adapted by including perceived support for ICT use (perceived ICT support) as a key component of the framework. This modified model was then employed to explore the relationships among Chinese EFL teachers' perceptions of the usefulness and ease of use of ICT, prevalence of ICT support, attitudes towards ICT, and intentions to adopt it in practice. Overall, the collected data provide substantial evidence for the credibility of the adjusted TAM and corroborate recent research on teachers' attitudes and ICT integration practices (e.g., Ertmer et al. 2012; Eze et al. 2021; Liu et al. 2017a; Lv et al. 2024; Moradi and Chen 2022; Tondeur et al. 2017), which found that teachers' perceived ICT support plays a significant role in influencing how effectively they incorporated ICT in their teaching. These outcomes suggest that incorporating perceived ICT support into teachers' TAM-centred investigations offers valuable insights and is a useful addition to this model. While our findings on ICT support align with those of previous studies, they reveal a notably stronger emphasis on institutional support compared to that found in Western contexts (e.g., Ertmer et al. 2012). This difference may stem from China's hierarchical educational structure and collective decision-making culture, where institutional support plays a more decisive role in technology adoption compared to individual initiative.

The strong correlation between perceived ICT support and ease of use ($r = 0.587$) exceeds that of similar relationships reported in several Western studies (e.g., Alonzo et al. 2024; Tondeur et al. 2017), suggesting that institutional support may be particularly crucial for Chinese teachers' confidence in using educational technology.

The role of ICT support in our findings reflects broader patterns in Asian educational systems, where institutional backing plays a more central role than in Western contexts. While studies from Europe and North America (e.g., Alonzo et al. 2024; Salokangas et al. 2019; Tondeur et al. 2017) often emphasise individual teacher initiative, our findings align with the characteristic features of Asian educational contexts, where institutional support structures and collective decision-making are crucial elements in educational innovation.

RQ2. The proliferation of ICT in education presents a range of potential advantages and obstacles for educational institutions at all levels, from early education to university (Moradi 2022). Teachers, policymakers, practitioners, and others dealing with education understand and acknowledge the importance and value of ICT in teaching and learning. However, the emergence of modern technology in education has made its adoption and incorporation more difficult for teachers.

The factors influencing ICT integration in English language classes can be divided into three major categories: (1) personal, (2) institutional, and (3) technological. Within the personal domain, teachers' attitudes towards ICT may affect its adoption. Specifically, language teachers with positive attitudes regarding the integration of contemporary educational technologies can easily demonstrate the insight and perceptiveness that contribute to technology acceptance and the perception of the usefulness of ICT adoption in teaching–learning processes.

At the institutional level, in line with Hu and Li (2018) and Yang et al. (2023), useful and advanced equipment, sufficient funding, appropriate training programmes, technical support, and technology accessibility in the classroom are the major factors that affect teachers and the useful and effective integration of ICT in teaching–learning processes. As Moradi (2014) explained, teachers' professional development constitutes a critical factor in the effective and successful integration of ICT into teaching practices. The abovementioned points, in a broader sense, may significantly affect language teachers' attitudes towards the implementation of ICT in their classes.

Another major factor influencing the successful integration of ICT in language education is directly related to the technological level; language teachers need to regard ICT in line with their previous teaching experiences and demands and consider it a valuable educational resource that can be accessed and used easily (Li et al. 2017; Chen et al. 2021; Yang et al. 2023). In particular, teachers must have opportunities to test and use modern technologies before fully integrating them into their teaching. Additionally, they need to be able to observe the outcomes of ICT adoption in the teaching–learning processes. In other words, language teachers need to feel confident that the adoption of ICTs in their classes will facilitate teaching–learning processes that will ultimately lead to a more interesting and effective teaching method for both students and teachers.

Our findings regarding these three categories of factors reveal both similarities and distinctive patterns compared to those of previous research. While personal factors align closely with international studies, the institutional factors in our study exhibited a stronger influence than that typically reported in Western contexts (e.g., Tondeur et al. 2017; Ertmer et al. 2012). This divergence likely reflects China's centralised educational system and the greater emphasis on collective implementation of educational innovations, a finding that aligns with other Chinese context studies (Hu and Li 2018).

While our findings provide valuable insights into ICT integration in Chinese EFL education, they should be interpreted within their specific regional context. Guangzhou, as one of China's most developed cities, represents an urban educational environment with relatively advanced technological infrastructure and institutional support. This context may differ from other regions in China, particularly in terms of resource availability and institutional capacity. However, the strong alignment of our findings with those of other Chinese studies (e.g., Liu et al. 2017a; Hu and Li 2018) suggests that the fundamental patterns we observed regarding institutional influence and collective implementation approaches may be representative of broader trends in Chinese higher education.

Limitations and future research directions

This study has several limitations. First, while the empirical validity of the TAM is well-documented, it has inherent constraints, particularly the disparity between individuals' intentions to adopt technology and their actual usage patterns (Bagozzi 2007). Previous TAM-based studies often prioritised intention as an outcome variable while overlooking its critical link to actual

behaviour (Wu and Du 2012). Meta-analyses have yielded mixed findings regarding this association. Turner et al. (2010) found that behavioural intention generally correlates with actual usage, whereas Wu and Du (2012) questioned the suitability of intention as a proxy for usage. Further empirical research is required to explore this relationship in diverse contexts. Second, attitudes play a pivotal role in shaping EFL teachers' willingness to integrate ICT into their classrooms. However, the relationship between attitudes and actual behaviours remains complex and inconsistent. Questions persist about why attitudes do not always predict behaviour and, in some cases, why their associations are weak (Karahan 2007; Luttrell and Sawicki 2020). Exploring these discrepancies can provide valuable insights into the dynamics of ICT adoption in educational settings.

This study's sample of 186 EFL teachers from tertiary institutions in Guangzhou offers both strengths and limitations regarding generalisability. As a first-tier city with advanced educational infrastructure, Guangzhou exemplifies optimal conditions for ICT integration in Chinese higher education. However, this setting may not fully represent the challenges faced by institutions in less developed regions or rural areas, where technological infrastructure, institutional support, and teacher training opportunities may differ significantly. Furthermore, while the findings align with broader trends in Chinese higher education, caution is needed when generalising these results to other educational levels or EFL teaching contexts in Asian countries with different institutional structures and cultural traditions.

Future research should address these limitations through several approaches. First, incorporating samples from diverse regions, especially less developed areas of China, could provide a more comprehensive understanding of ICT integration across varying socioeconomic contexts. Second, comparative studies among Chinese cities of differing tiers would elucidate how technological integration patterns correlate with regional development. Third, cross-regional analyses within China could shed light on how institutional and technological factors vary. Finally, expanding the research scope to other Asian educational systems with similar cultural and institutional characteristics would offer insights into the broader applicability of these findings and advance our understanding of ICT integration in Asian contexts.

Conclusions

This study investigated the factors influencing ICT adoption in language teaching and learning at the higher education level with a focus on EFL teachers' perceptions of ICT integration and its contributions to English language teaching. The findings indicate that all of the influencing factors are interrelated. The correlations among the TAM variables suggest that perceived usefulness, attitudes towards ICT, and perceived ease of use are strong predictors of intention to adopt ICT. ICT support also significantly predicts ease of use and technological competence. Teachers' experience with technology, along with their attitudes and perceptions, impact both their intention and actual ICT use in the teaching-learning processes. These findings affirm that the revised TAM framework is a robust tool for explaining EFL teachers' ICT implementation in diverse educational settings.

In the context of Chinese higher education, particularly in first-tier cities like Guangzhou, these findings underscore the importance of institutional support and training programmes to enhance teachers' ICT skills and perceptions of its usefulness. Such efforts are likely to foster greater adoption and effective pedagogical integration of ICT in EFL teaching. Participants' positive responses regarding ICT's impacts confirm that professional development programmes are crucial for accelerating ICT integration and helping build confident and skilled EFL teachers.

However, many teachers perceive ICT integration as a challenging task requiring sustained pedagogical and technical support.

Based on these findings, we propose several recommendations for policymakers and institutions. At the professional development level, institutions should design comprehensive training programmes that progress systematically from foundational ICT skills to advanced pedagogical applications. These should include hands-on workshops tailored to EFL contexts supported by peer mentoring systems where experienced ICT users guide their colleagues. Institutions should also establish dedicated ICT support teams, provide online repositories of teaching resources and best practices, and ensure real-time technical assistance during classroom implementation. Regular feedback mechanisms between teachers and IT support staff are essential, as is allocating time within teachers' schedules for ICT training and practice. Incentive systems recognising innovative ICT use, along with clear guidelines for integrating ICT into the EFL curriculum, would further encourage adoption. Additionally, developing assessment frameworks to evaluate ICT integration's effectiveness would ensure continuous improvement.

Modern technologies are transformative tools for language teachers that enable more effective and dynamic teaching and learning. ICT integration in English language teaching can potentially enrich the curriculum and enhance student engagement, creativity, problem-solving skills, self-confidence, collaboration, communication, and digital literacy. However, the effectiveness of ICT integration varies across institutional contexts and regions within China, particularly between urban and rural areas.

To address these regional disparities, institutions should adopt context-specific implementation strategies that consider local resources and constraints. Flexible support systems tailored to different institutional capabilities are crucial. Establishing regional networks for resource sharing and fostering partnerships between well-resourced and under-resourced institutions can help bridge the digital divide and promote equitable ICT integration across diverse educational settings. Systematic ICT integration has the potential to transform teaching and learning practices. However, its impacts and implementation strategies must be adapted to align with local contexts and resources, ensuring relevance and effectiveness.

Data availability

The data are available from the corresponding author on reasonable request.

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Author contributions

The author, HM, is solely responsible for the conception, design, execution, and analysis of this research, as well as the preparation of the resulting manuscript.

Competing interests

The author declares no competing interests.

Ethical approval

Approval was obtained from the academic committee of the School of Foreign Studies (SFS), Guangzhou University (No. 2023092001), on 20.09.2023. The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

Informed consent

Written informed consent was obtained from all participants before their participation in the study. Participants were explicitly informed about the purpose of the research, measures taken to protect their anonymity, the intended use of their data, and any potential risks associated with participation. Participation in the study was voluntary. Data privacy and confidentiality were safeguarded at every stage of data collection in strict compliance with established protocols and ethical standards.

Additional information

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